

SecureMailScope Forensic Investigation Report

SIH26159 | National Technical Research Organisation (NTRO)

Investigation ID	INV-6D9C84A0	Security Posture	70.0/100 (MEDIUM RISK)
PCAP Artifact	INV-6D9C84A0_wireshark_real_smtp.pcap	Completeness	80.0%
SHA-256	17ad230db1b6fd5dd18eb311092d...	Generated	2026-09-18T11:58:08

AI-Assisted Investigation Summary & Reasoning

AI-Assisted Synthesis • Strictly grounded in deterministic evidence ledger records

Generated by: nvidia (meta/llama-3.2-11b-vision-instruct) | Prompt Version: 2.1.0

Executive Briefing: Medium risk level detected due to cleartext authentication transmission and deprecated TLS session. Recommended actions include implementing encryption and upgrading to a more secure TLS version.

Risk Profile Assessment: The presence of cleartext authentication transmission and a deprecated TLS session increases the risk level to medium. This is because cleartext authentication transmission exposes user credentials to potential eavesdropping, while the deprecated TLS session indicates a potential vulnerability to attacks.

Verified Forensic Findings

Cleartext Authentication Transmitted [HIGH]

User credentials (LOGIN/AUTH/USER/PASS) were observed unencrypted across the network wire.

Remediation: Prohibit cleartext authentication over unencrypted ports. Require explicit STARTTLS before AUTH.

Supporting Evidence: E-9CDF83, E-8B46A8, E-10AE45, E-F03E56, E-A7D7F7

Evidence Ledger Excerpt

ID	Type	Observed Claim	Source
E-9CDF83	OBSERVED	PCAP capture 'INV-6D9C84A0_wireshark_real_smtp.pcap' loaded:...	CaptureEngine
E-8B46A8	OBSERVED	Discovered 1 TCP stream(s) with protocols: SMTP....	TCPReconstructionEngine
E-10AE45	starttls_advertised	SMTP Server advertised STARTTLS capability in EHLO response....	SMTPAnalyzer
E-F03E56	OBSERVED	PCAP capture 'INV-6D9C84A0_wireshark_real_smtp.pcap' loaded:...	CaptureEngine
E-A7D7F7	OBSERVED	Discovered 1 TCP stream(s) with protocols: SMTP....	TCPReconstructionEngine
E-6A0386	starttls_advertised	SMTP Server advertised STARTTLS capability in EHLO response....	SMTPAnalyzer
E-75BB18	crypto_rule_triggered	Rule Triggered [RULE-CLEARTEXT-AUTH]: Cleartext Authenticati...	CryptoRuleEngine

E-F311B3	ML_CLASSIFICATION	XGBoost Classifier classified session as 'DEPRECATED_TLS' (R...	CryptoRiskClassifier (XGBoost)
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